

APPLICATION OF MACHINE LEARNING IN HEALTHCARE: PREDICTIVE GRADING OF GLIOMAS

Final Project for Module 12: Advanced Machine Learning in Health
Care (AMLHC)

Can the malignancy grades of gliomas (LGG vs. GBM)
be precisely predicted using k-NN and artificial neural
networks?

Submitted by: Emilia Schloß

UNIT Tirol

12.06.2026

KDD PIPELINE – DATA STRUCTURE & MEANING OF VALUES

Dataset Overview (Glioma Grading Dataset):

- ▶ Target variable (y): Grade -> Binary classification: 0 (Low-Grade Glioma) vs. 1 (Glioblastoma / High-Grade Glioma)
- ▶ Clinical features: Age_at_diagnosis (metric), Gender & Race (categorical)
- ▶ Molecular features: 20 specific gene mutations (e.g., IDH1, TP53, ATRX).

Methodological core: 0 vs. NaN (the "Important Hint"):

- ▶ NaN (Not a Number): Represents incomplete data or errors during data collection.
- ▶ 0 (zero value): Represents genuine, critical information in this project.
- ▶ Biological significance: 0 = Wild type / No mutation present.
- ▶ 1 = Gene is mutated.
- ▶ Preprocessing check result: The dataset contains 0 genuine missing values (NaN). All zero values are incorporated into the models as biological facts.

KDD PIPELINE – STATISTICAL PREREQUISITES

Analysis of age (Age_at_diagnosis):

- ▶ Prerequisite check: Shapiro-Wilk test for normal distribution
- ▶ Result: $p\text{-value} < 0.05 \rightarrow$ Age is not normally distributed
- ▶ Methodological consequence: Exclusion of the parametric Pearson coefficient. Use of non-parametric Spearman rank correlation for feature selection.

Analysis of mutations (binary features, 0/1):

- ▶ The mathematical problem: Mutations are nominal/binary in nature; therefore, Pearson and t-tests are inappropriate, as they require continuous, normally distributed data.
- ▶ Methodological solution: Use of the chi-square test of independence (χ^2).
- ▶ Objective: Identification of genes showing a statistically significant association ($p < 0.05$) with tumor grade.

MODEL 1 – K-NEAREST NEIGHBORS (K-NN)

Model characteristics:

- ▶ Approach: Distance-based instance learner (lazy learning) -> No explicit model-building phase occurs during training
- ▶ Parameter selection: $k=5$ (an odd number was chosen to rule out a tie in the majority vote for the binary classification of LGG vs. GBM)
- ▶ Distance metric: Euclidean distance for determining nearest neighbors

Justification of Choices:

- ▶ Problem: The `Age\_at\_diagnosis` feature has a value range of approximately 20 to 90 years, whereas the mutation data (genes) are purely binary (0 or 1).
- ▶ Consequence without scaling: Age would mathematically dominate the Euclidean distance calculation entirely, rendering the biological information from the gene mutations ineffective for the model.
- ▶ Solution: Mandatory standardization (StandardScaler) prior to modeling to ensure all features are weighted equally.

MODEL 2 – ARTIFICIAL NEURAL NETWORK (ANN)

Model characteristics:

- ▶ Approach: Function-based classifier (Multi-Layer Perceptron / MLP)
- ▶ Strength: Can automatically learn complex, non-linear relationships between clinical factors and gene mutations
- ▶ Challenge: High computational cost compared to k-NN and lack of direct interpretability ("black box")

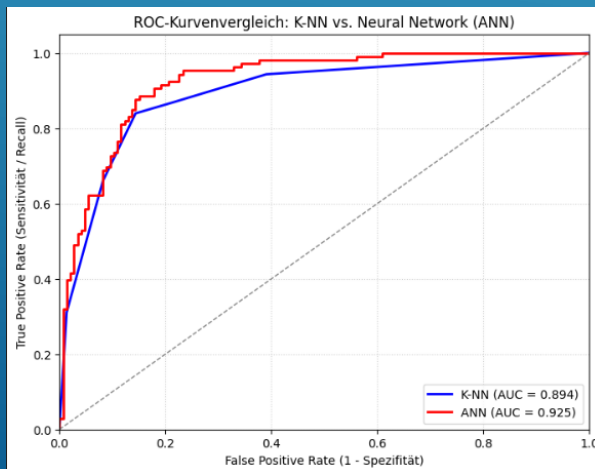
Network architecture selected for the project:

- ▶ Input layer: Represents the number of statistically significant features (age + relevant gene mutations)
- ▶ Hidden layer: 1 layer with 16 neurons (enables the learning of feature combinations)
- ▶ Output layer: 1 neuron with binary output (0 for LLG, 1 for GBM)
- ▶ Optimization: Max. 500 iterations (epochs) to ensure complete weight convergence via backpropagation

RESULTS – VALIDATION & PERFORMANCE METRICS

Model comparison at a glance (test set & 5-fold CV):

Metric (Class 1 / GBM)	Model 1: K-NN	Model 2: ANN (MLP)
Precision	81,0%	81,0%
Recall	84,0%	85,0%
Testset	0,8200	0,8300
5-Fold VC	0,8308	0,8418
Classification errors (total)	38 patients	37 patients



Beyond Accuracy

- ▶ The problem: The dataset exhibits an unbalanced class distribution (class skew). Optimizing solely for accuracy would be mathematically misleading.
- ▶ The solution: Focus on recall (how many actual glioblastomas were detected in time?) and the F1-score as the harmonic mean.
- ▶ Cross-validation: Use of stratified 5-fold cross-validation to ensure an unbiased estimate of generalization performance across unbalanced folds.

INTERPRETATION & AI ETHICS (AI ETHICS AND BIAS)

The "Black Box" Conflict in Clinical Practice:

- ▶ ANN (Neural Network): Achieves the best overall performance (higher recall of 85.0%) but functions as a "black box." Its complex internal weights are not directly interpretable by medical professionals
- ▶ K-NN (Nearest Neighbor Search): Is mathematically slightly inferior (recall of 84.0%) but offers excellent transparency (Explainable AI). Decisions are based directly on the *k* most similar historical patient profiles
- ▶ Implication: In medicine, a lack of transparency is a significant issue. Physicians require systems with interpretable logic to assume ultimate responsibility for aggressive cancer therapies

Risk of Misclassification & Focus on Recall:

- ▶ A false negative (where a glioblastoma is erroneously classified as a low-grade glioma) delays vital, aggressive therapy
- ▶ Consequently, maximizing recall for the high-risk class takes precedence over overall accuracy in a healthcare context

Data Bias & Fairness (TCGA Dataset):

- ▶ The dataset is based on The Cancer Genome Atlas project
- ▶ Critical assessment of representativeness regarding demographic characteristics (race, gender) is required; the underrepresentation of certain ethnic groups limits fair generalizability in diverse, real-world clinical practice

INTERPRETATION & KI-ETHIK (AI ETHICS & BIAS)

Der „Black Box“-Konflikt im Klinikalltag:

- ▶ **ANN (Neuronales Netz):** Erzielt die beste Gesamtleistung (höherer Recall von 85,0 %), ist jedoch eine *Black Box*. Die verschachtelten Gewichte sind für Mediziner nicht direkt nachvollziehbar
- ▶ **K-NN (Nachbarschaftssuche):** Ist mathematisch leicht unterlegen (Recall 84,0 %), besitzt aber eine exzellente **Transparenz** (Explainable AI). Entscheidungen basieren direkt auf den k ähnlichsten historischen Patientenprofilen
- ▶ **Konsequenz:** In der Medizin wiegt der Mangel an Transparenz schwer. Ärzte benötigen nachvollziehbare Systeme, um die Letztverantwortung für aggressive Krebstherapien zu übernehmen

Fehlklassifikations-Risiko & Recall-Fokus:

- ▶ Ein *False Negative* (Glioblastom wird fälschlicherweise als Low-Grade Gliom eingestuft) verzögert eine lebensnotwendige, aggressive Therapie
- ▶ Die Maximierung des **Recalls** der Risikoklasse hat im Gesundheitswesen somit oberste Priorität vor der reinen Gesamtgenauigkeit (*Accuracy*)

Daten-Bias & Fairness (TCGA-Datenbasis):

- ▶ Der Datensatz basiert auf dem Projekt *The Cancer Genome Atlas*
- ▶ Kritische Prüfung der Repräsentativität bezüglich demografischer Merkmale (Race, Gender). Eine Unterrepräsentation bestimmter ethnischer Gruppen schränkt die faire Generalisierungsfähigkeit im realen, diversen Klinikalltag ein